



Lecture 21:

VLSI High-Speed I/O Receiver Circuits

Contents



This Topic: VLSI High-Speed I/O Receivers Circuits

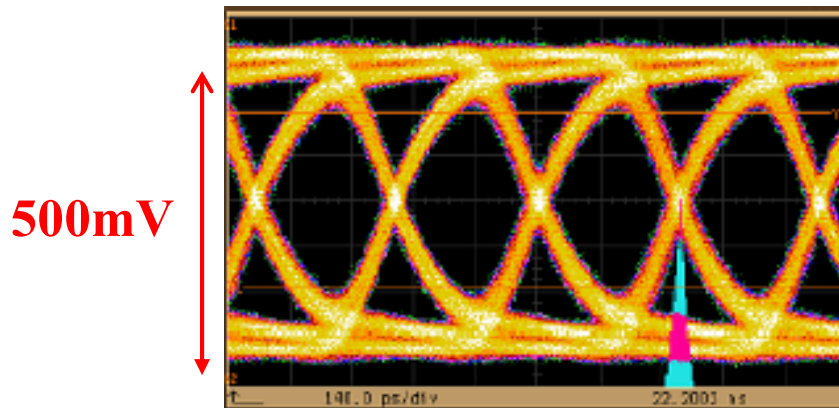
- Introduction
- Basic CMOS Receiver Circuit
- Improvements to Basic CMOS Receiver Circuit
- Other CMOS Receiver Circuit Types
- Design Examples

Introduction

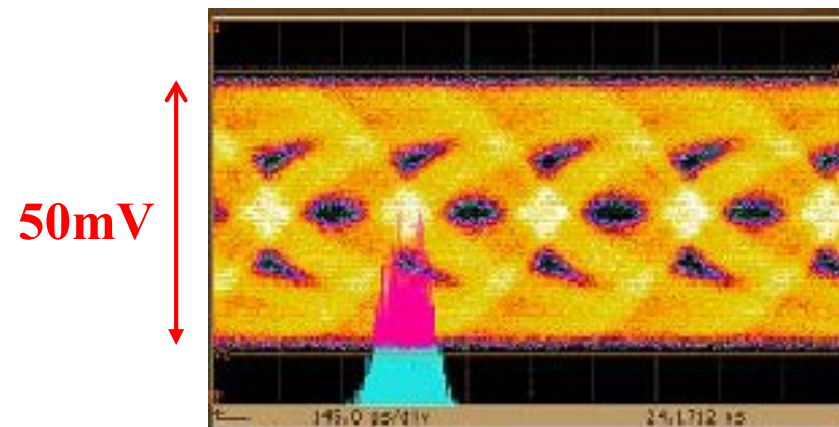


- Rx input signal is highly attenuated by high-speed I/O channel (e.g. $-15 \sim -20\text{dB}@5\text{Gh/z}$)
- Small input signal magnitude makes it very sensitive to voltage noises (ISI, x-talk, offsets, supply, reference, etc.)
- Severe Rx input signal eye width degradation makes it very sensitive to Rx sampling clock phase (skew and jitter) errors (RJ, DJ, I/Q phase error, etc.)

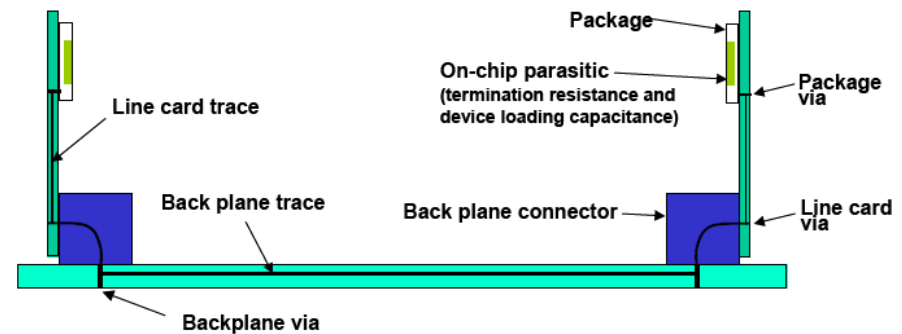
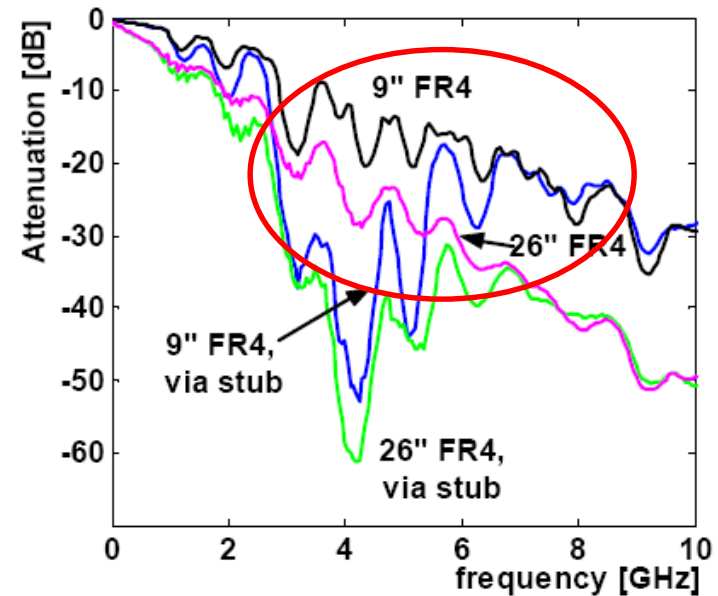
Introduction: Rx Input Signal Eye



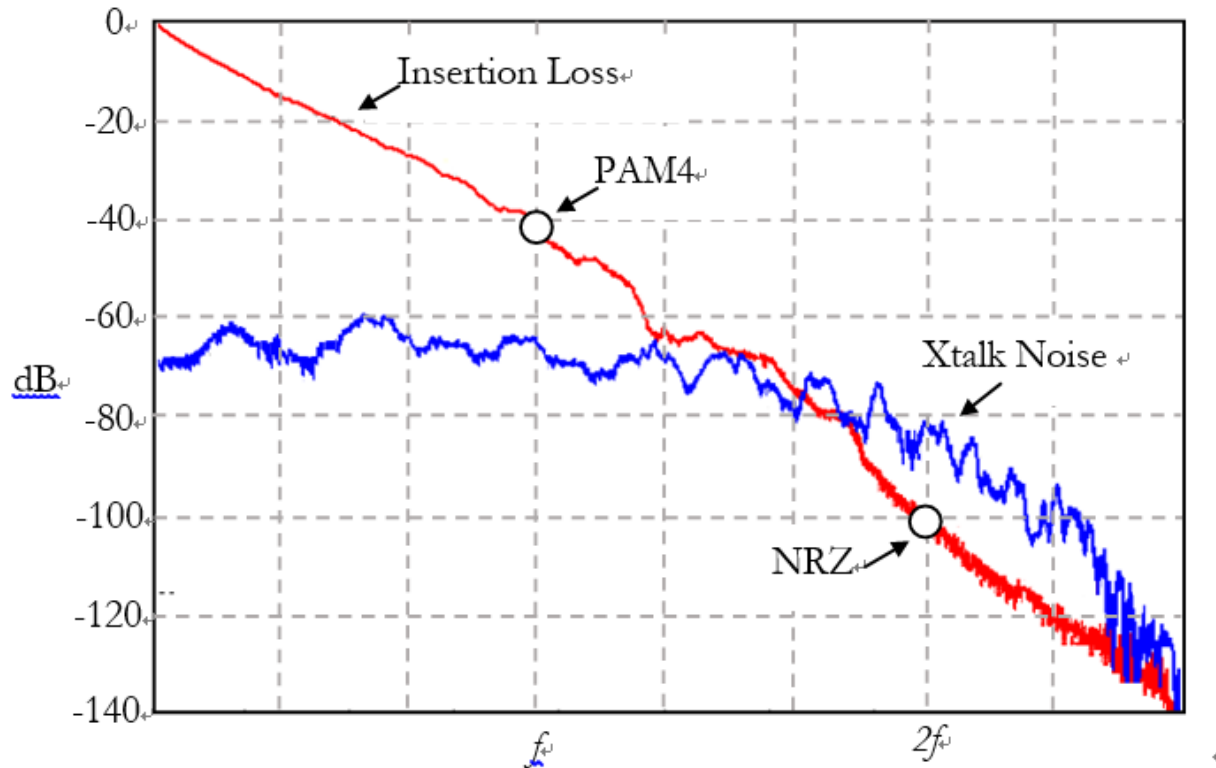
Tx output



Rx input



PAM-4 RX Motivation

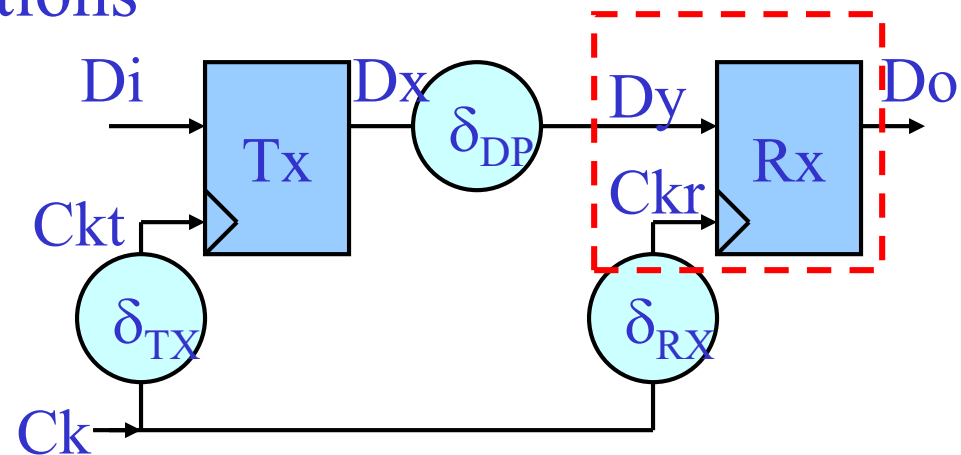


- Time domain signal may have low SNR at very high frequency
- Code some information in voltage domain

Introduction: Key Rx Design Considerations

- Key Rx design considerations

- BW limitation (ISI)
- X-talk
- Reflections/ Ringing
- Offset
- Supply/reference Noise
- Power/ESD
- etc.



- Rx need to be very sensitive to the Signal and not to the Noise !!

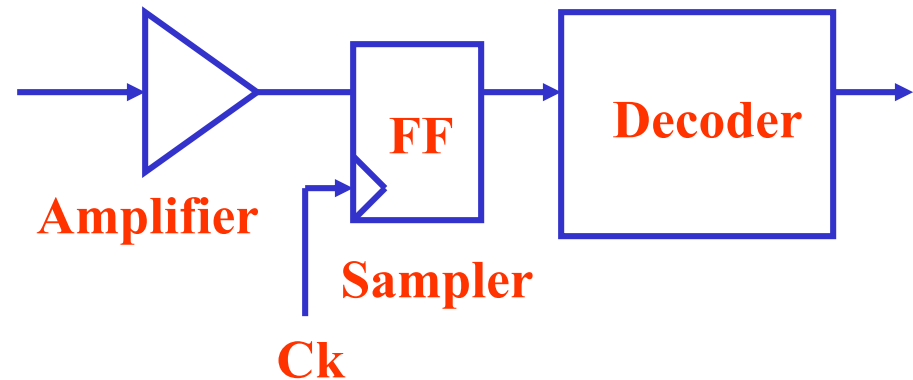
$$t_{hold} < \delta_{DP} + \delta_{Tx} - \delta_{Rx} < T - t_{setup}$$

Setup/Hold Times

Basic VLSI Rx AFE Circuit Elements

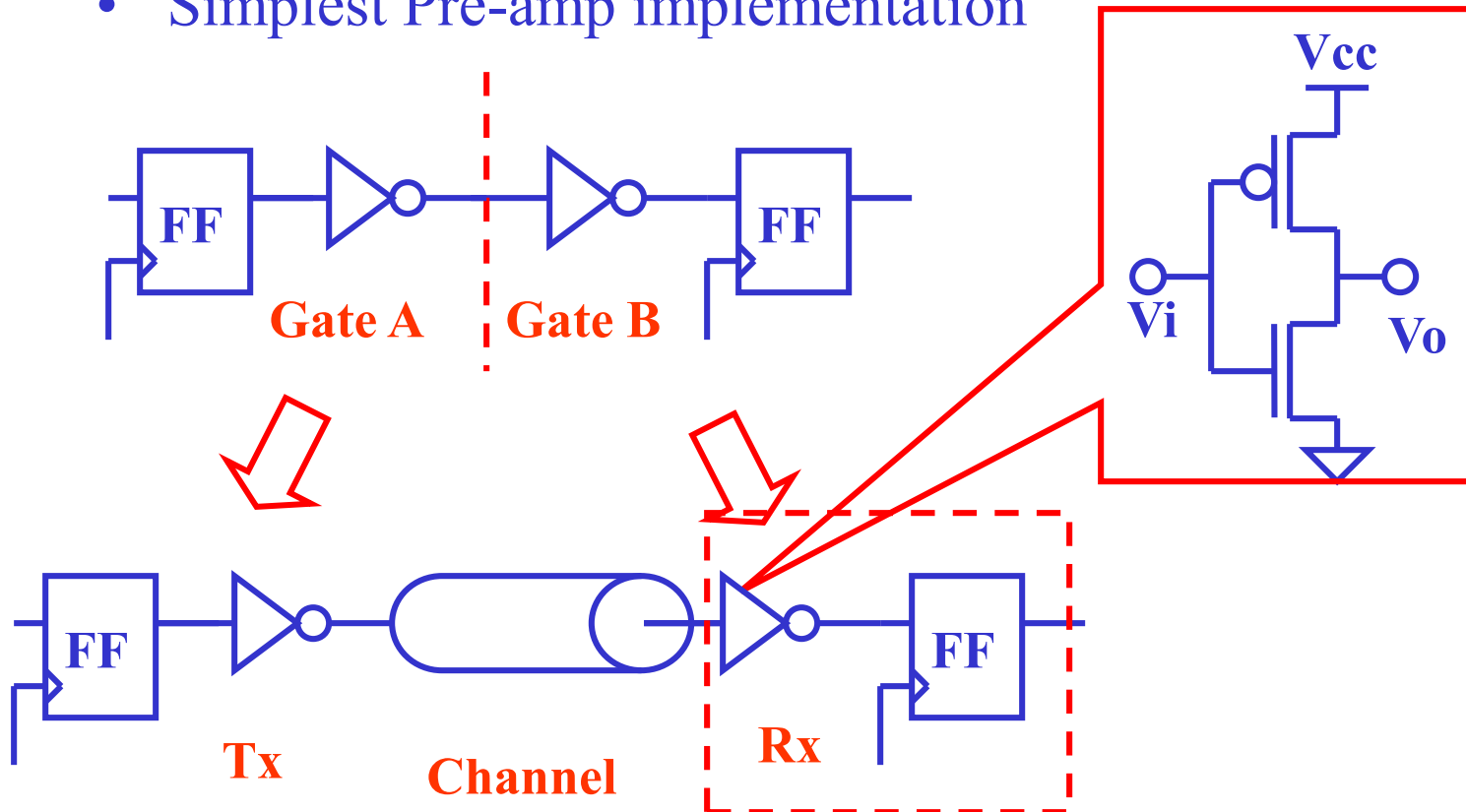


- Pre-amplifier
 - Signal amplification w/o or at low S/N penalty
- Sampler
 - Data recovery (A/D)
- Decoder
- Other features
 - R-Term,
 - Equalization,
 - ESD,
 - Offset Cal.
 - Squelch (power detection)
 - etc.



Pre-Amp Circuits: Basic Structure

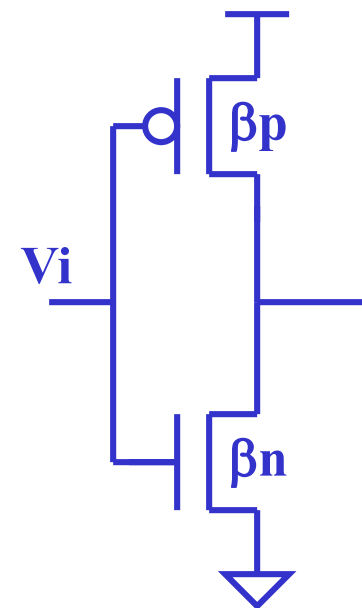
- CMOS gate based pre-amp is a “natural” transition from on-chip circuit to I/O circuit in VLSI technologies
- Simplest Pre-amp implementation



Key Limitations of Basic CMOS Pre-Amp

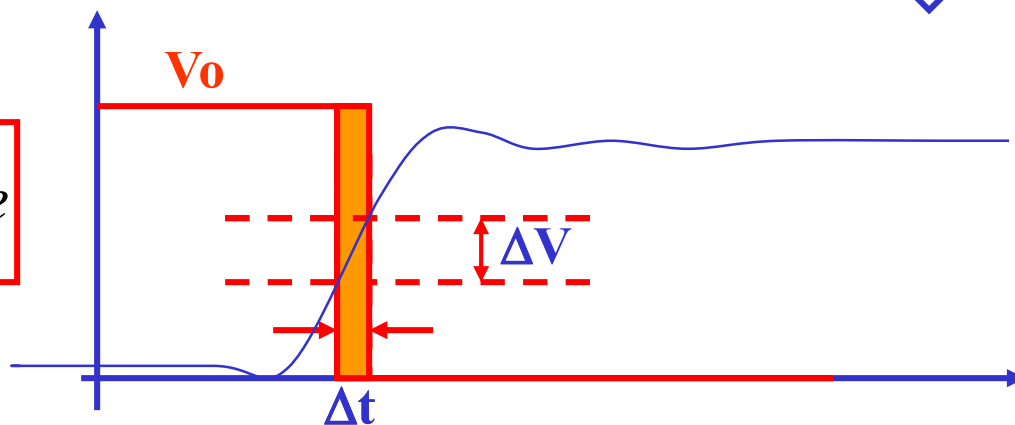


- High PVT sensitivity
 - V_{iH} - V_{iL} window of CMOS gate is PVT sensitive
- High jitter sensitivity
 - Single-ended operation
- High operation noise
 - Operation current not regulated
- Power dissipation highly PVT sensitive

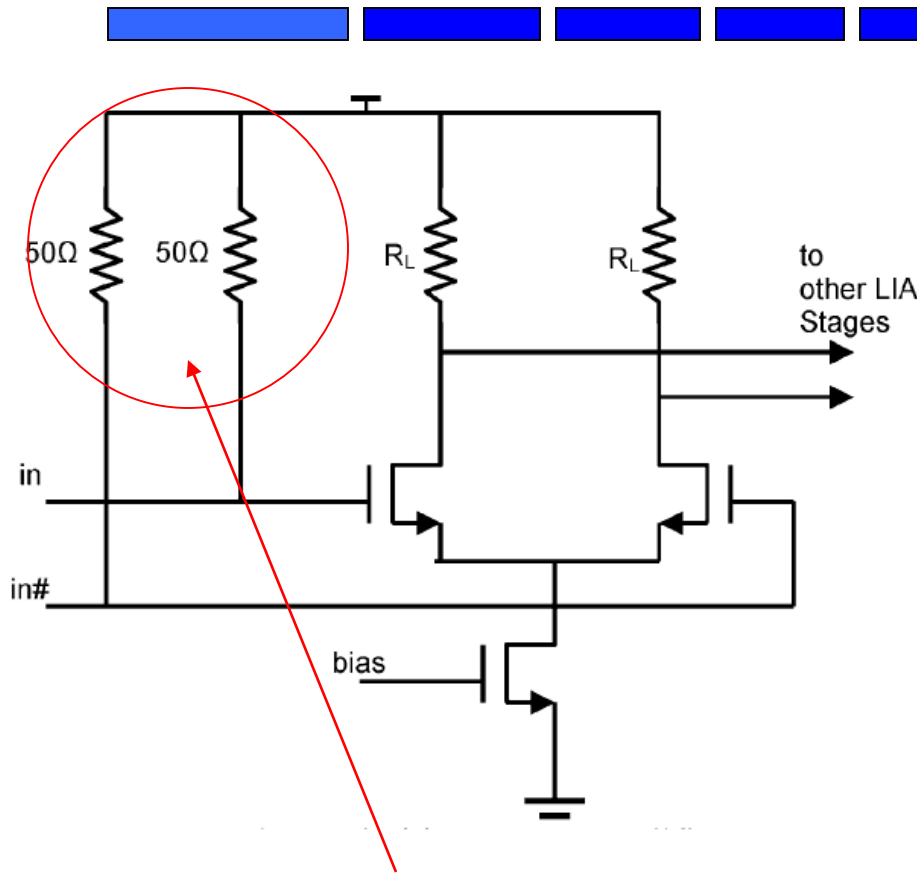


$$\Delta t = \Delta(V_i - V_{TH}) / Slope$$

Not correlated!!



Rx Pre-Amp Circuits: Differential Circuit

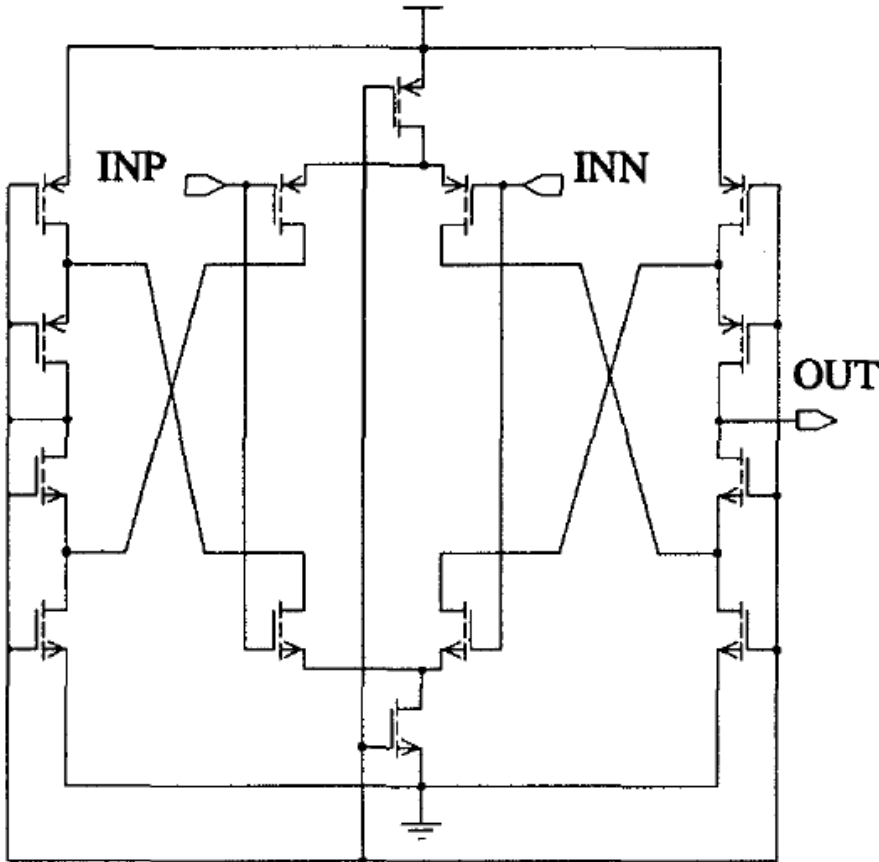


- Diff Pair input for lower noise
- NMOS input for high side signal
- Tail current bias for better PSRR
- Passive R load for better linearity
- R-Term for better signal quality

$$\text{Gain} = R_L * G_m$$

- R-Term to Vcc
- Input Biasing

VLSI Rx Pre-Amp Circuit Implementations



- Diff Pair input for lower noise
- CMOS input for wide V_{CM} signal
- Self-biasing for design simplicity
- Diff to single-end conversion

VLSI Rx Pre-Amp Circuit Implementations

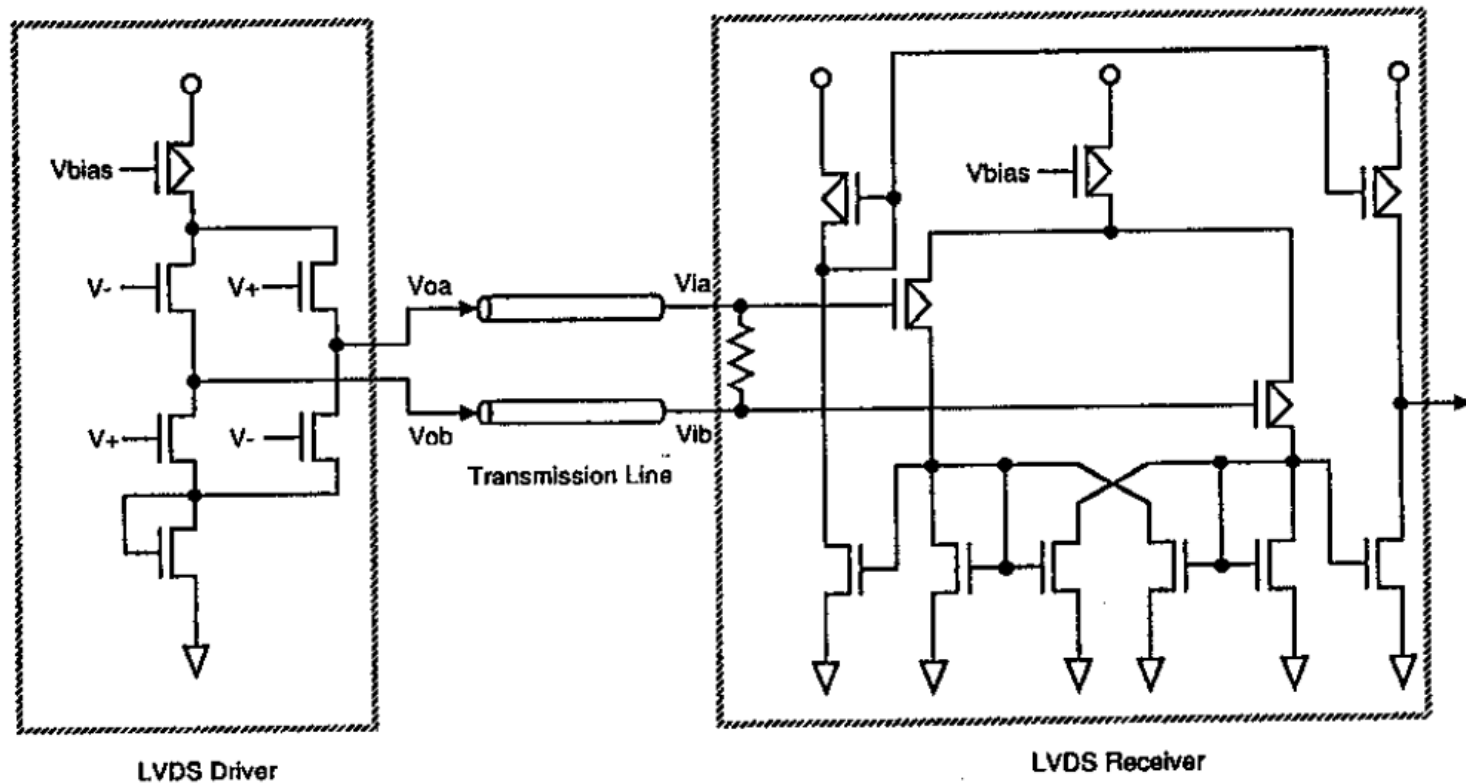
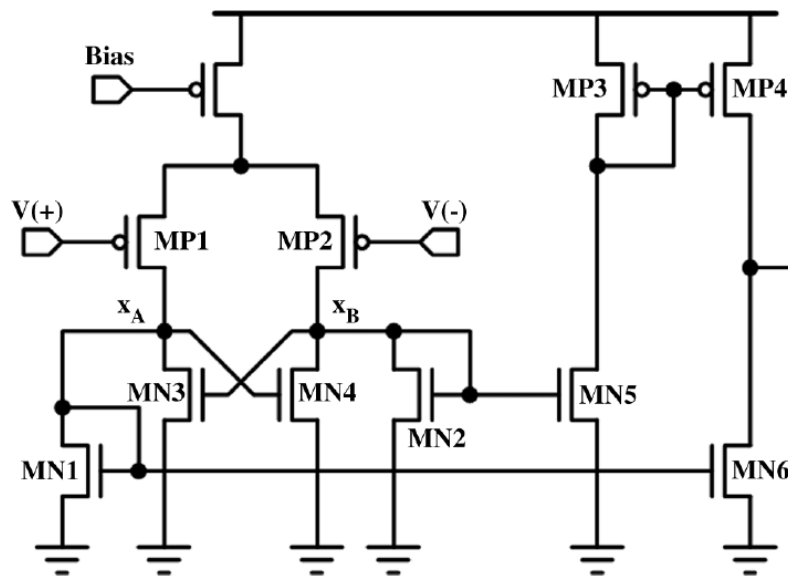


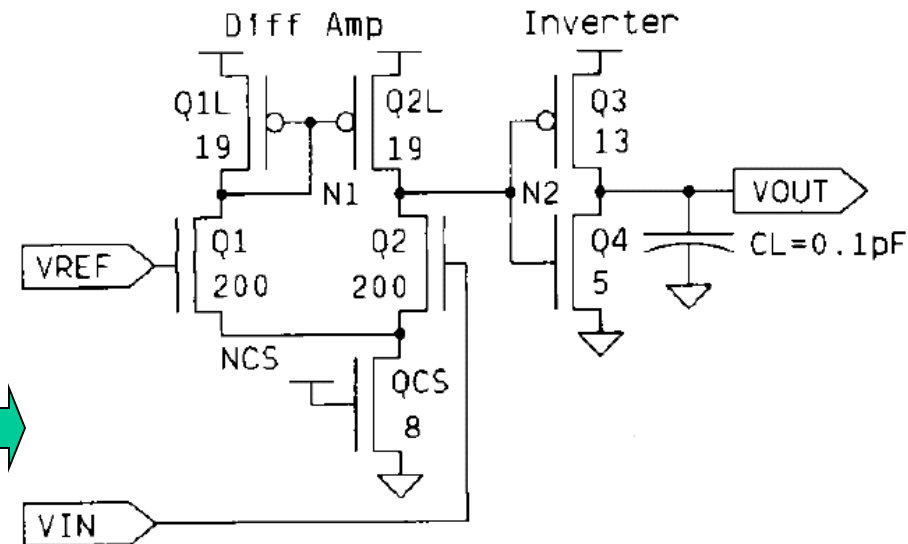
Figure 3: LVDS driver and receiver schematic.

VLSI Rx Pre-Amp Circuit Implementations

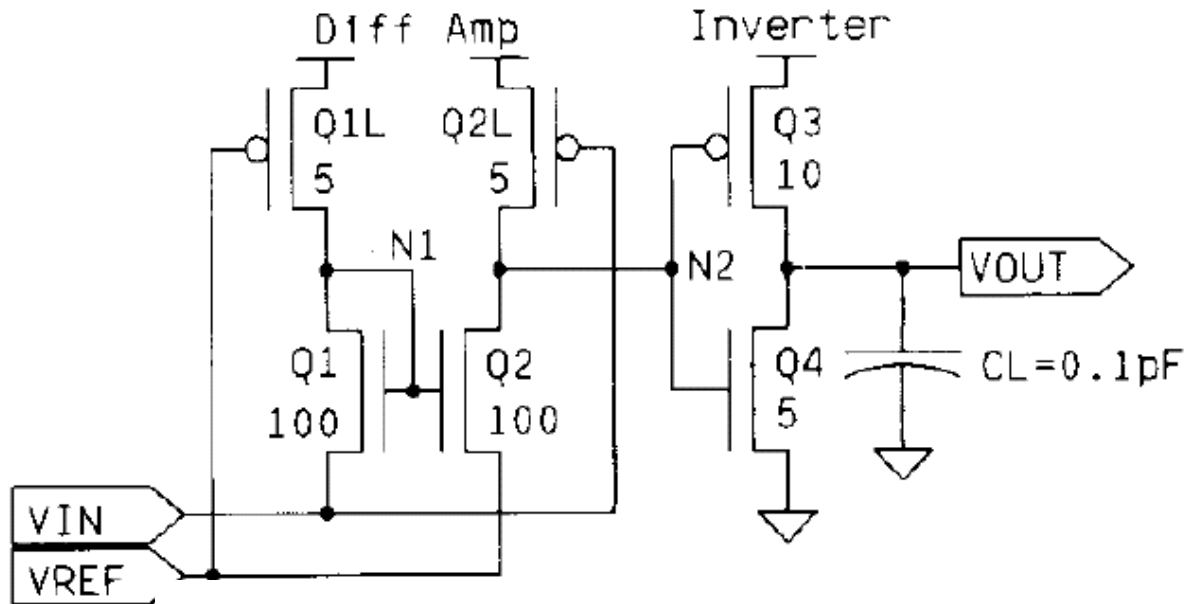


- PMOS input for low side signal
- Tail current bias for better PSRR
- Negative R for gain-boosting
- Diff to Single-end conversion

- NMOS input for high side signal
- Tail current bias for better PSRR
- Diff to Single-end conversion

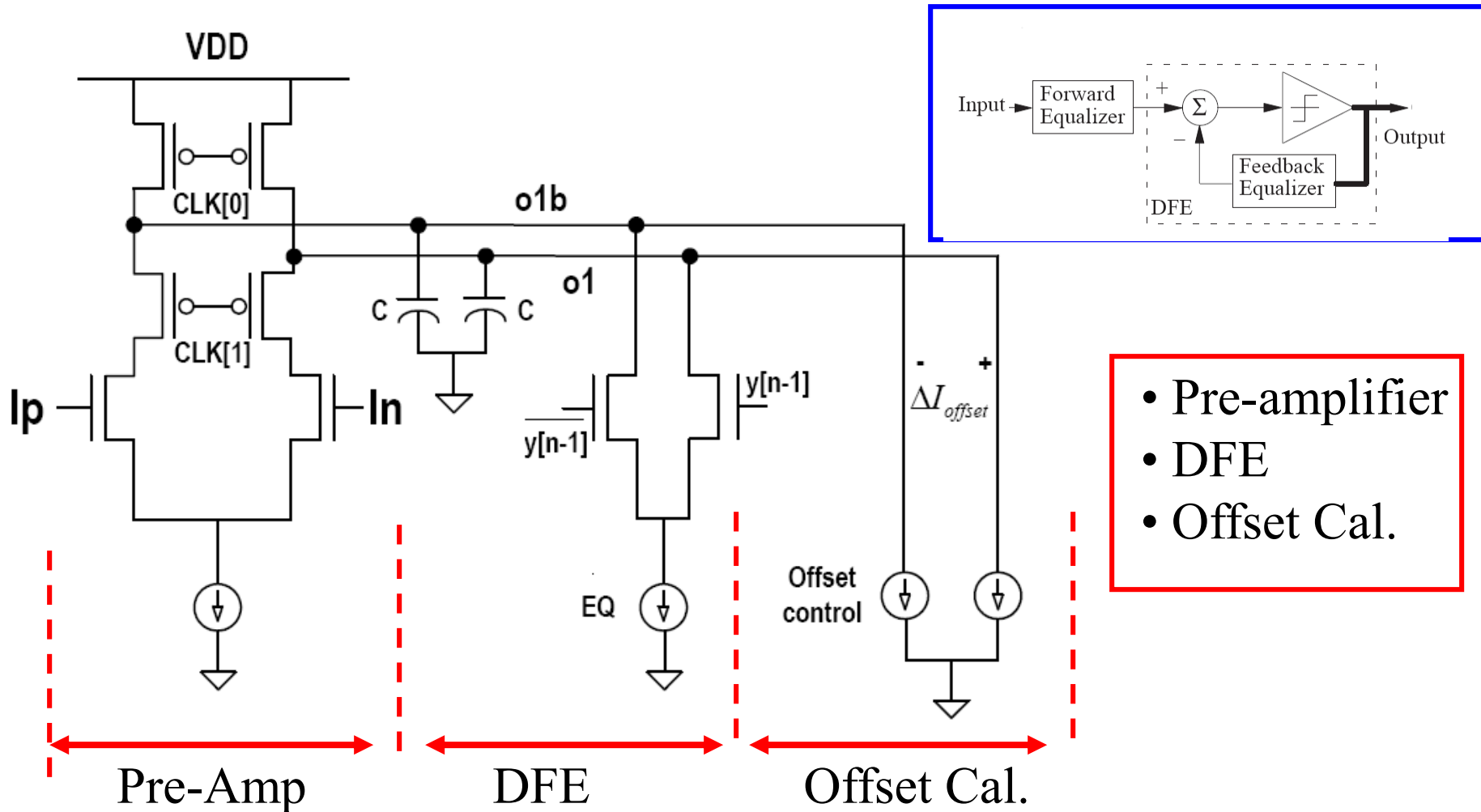


VLSI Rx Pre-Amp Circuit Implementations

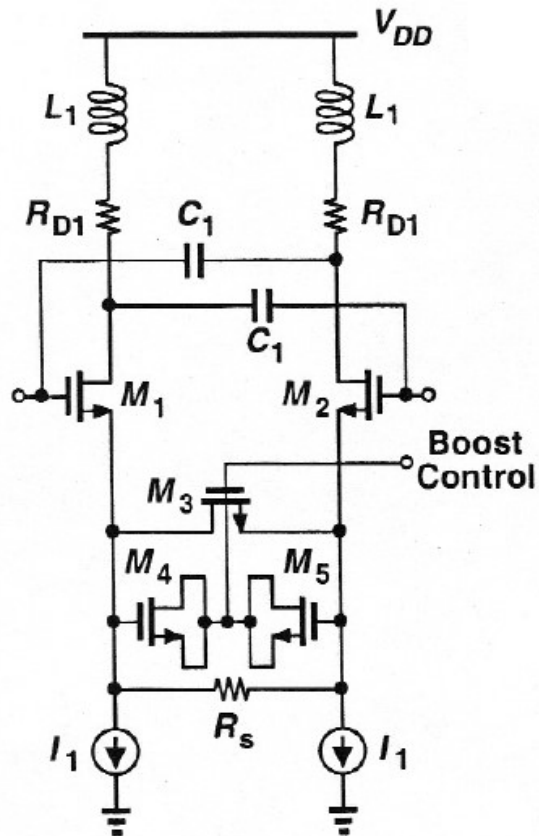


- Low side signal
- Source input for R-Term
- Pseudo Differential

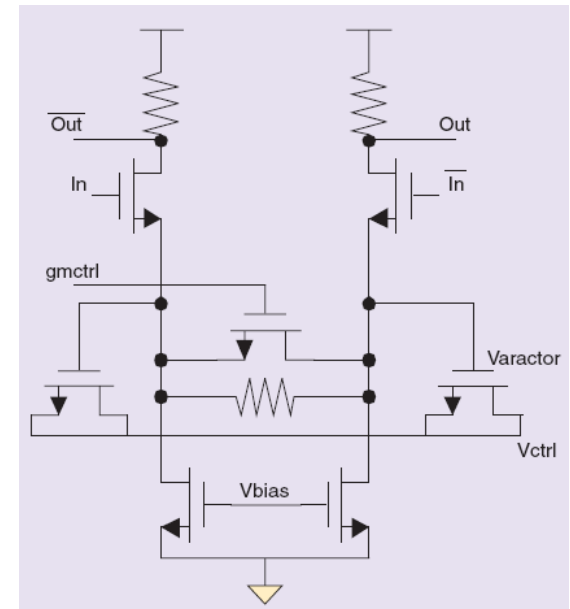
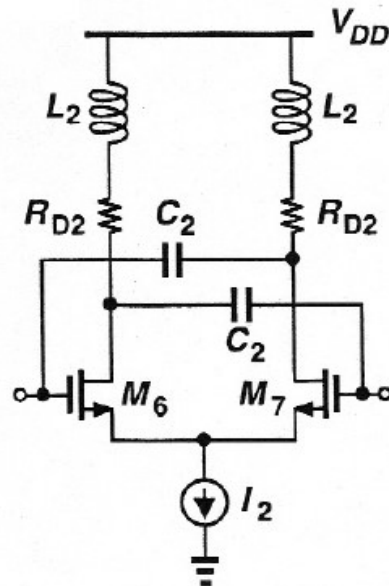
VLSI Rx Pre-Amp Circuit Implementations



VLSI Rx Pre-Amp w/ Equalization

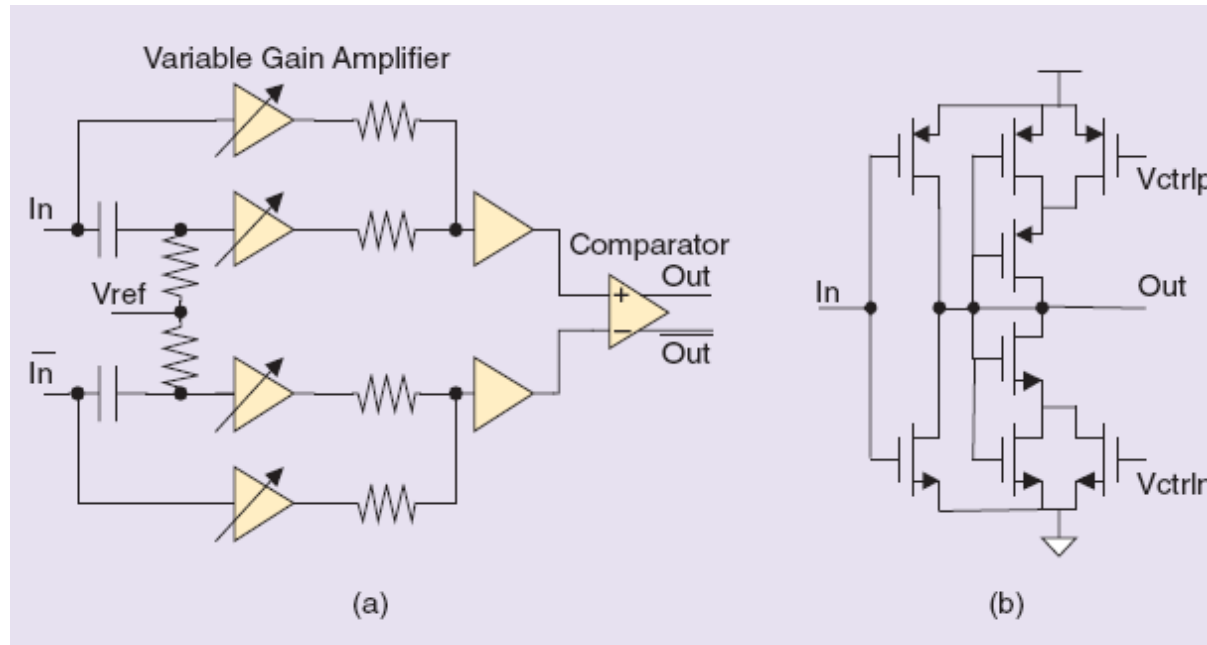


- Inductive load peaking
- Source Deg. FFE

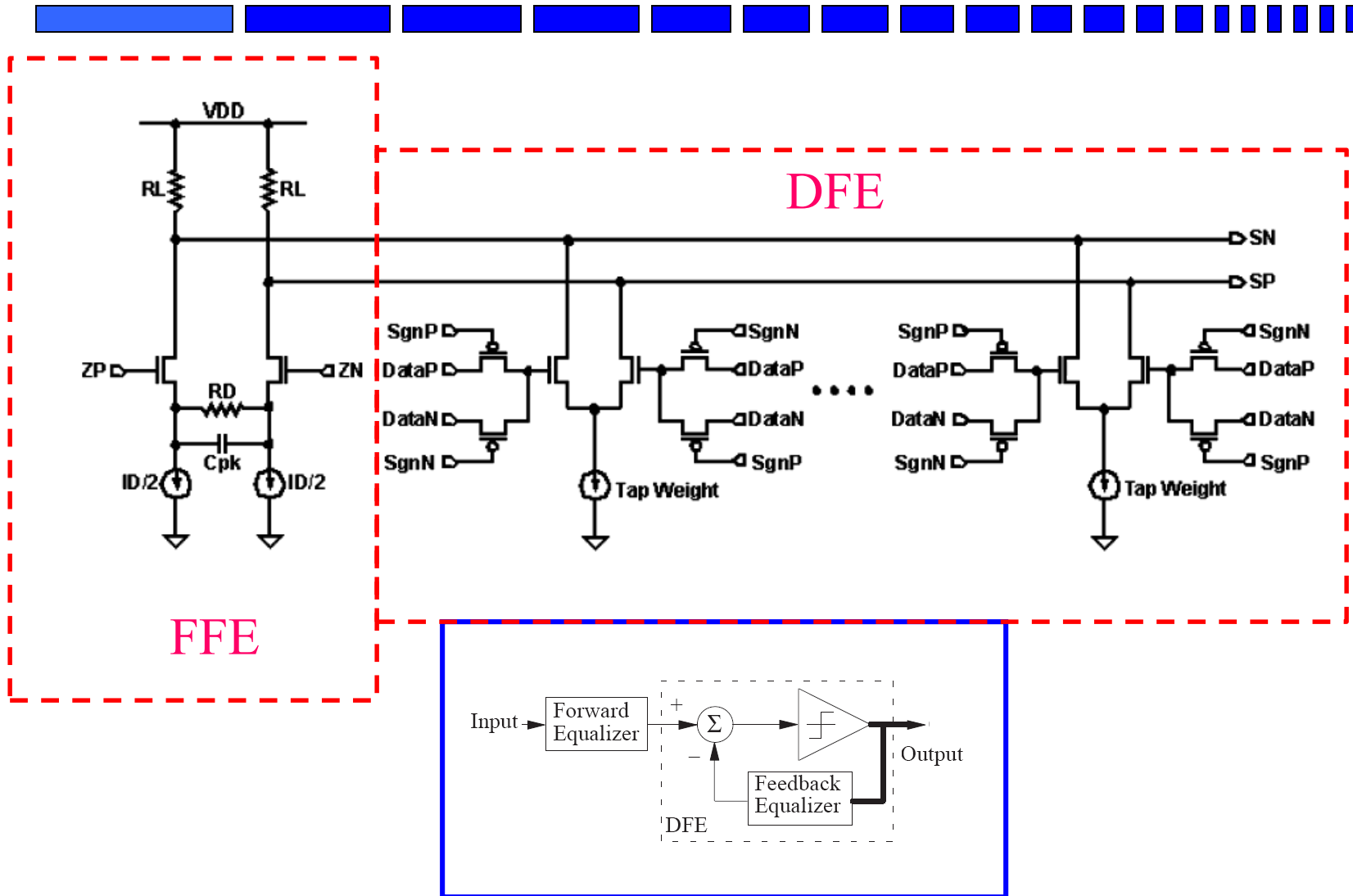


- Pre-Amp
- Source Deg. FFE

Rx Pre-Amp w/ Equalization



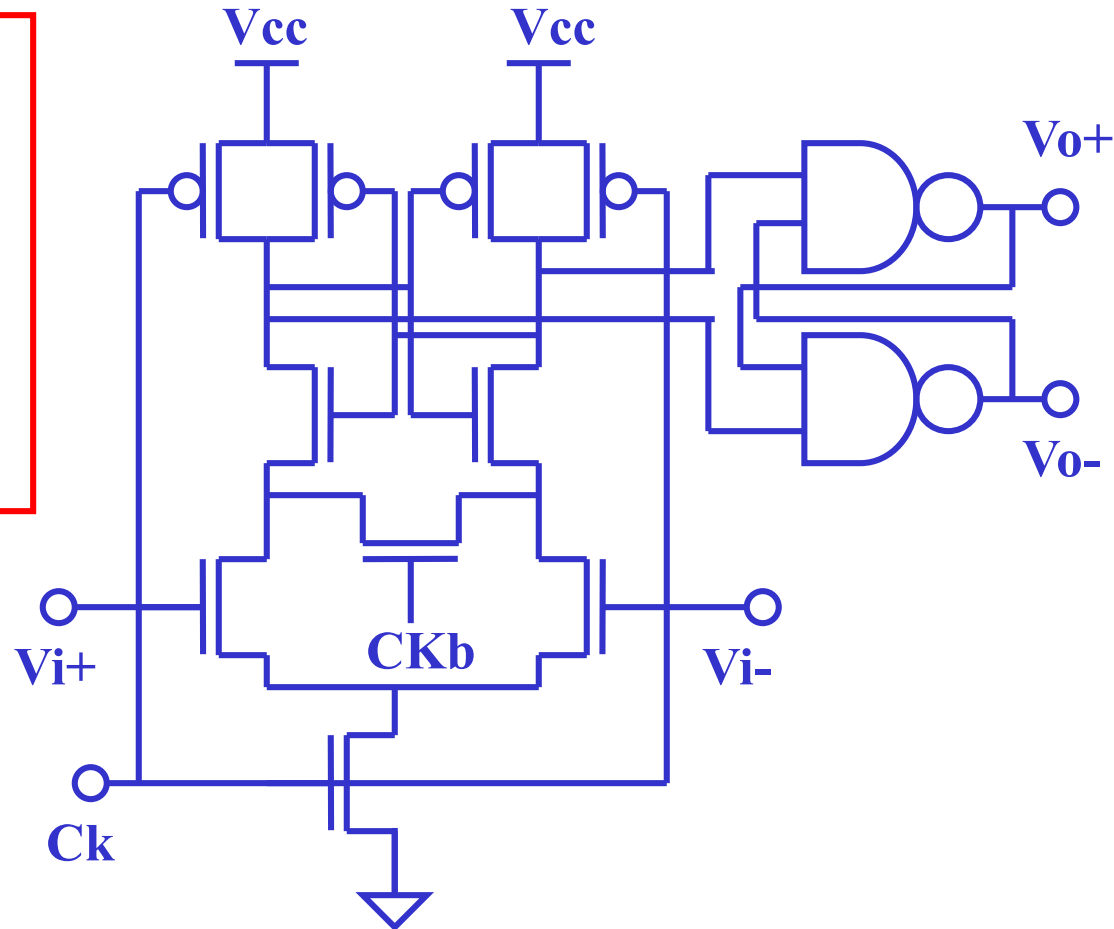
Rx Pre-Amp with FFE & DFE



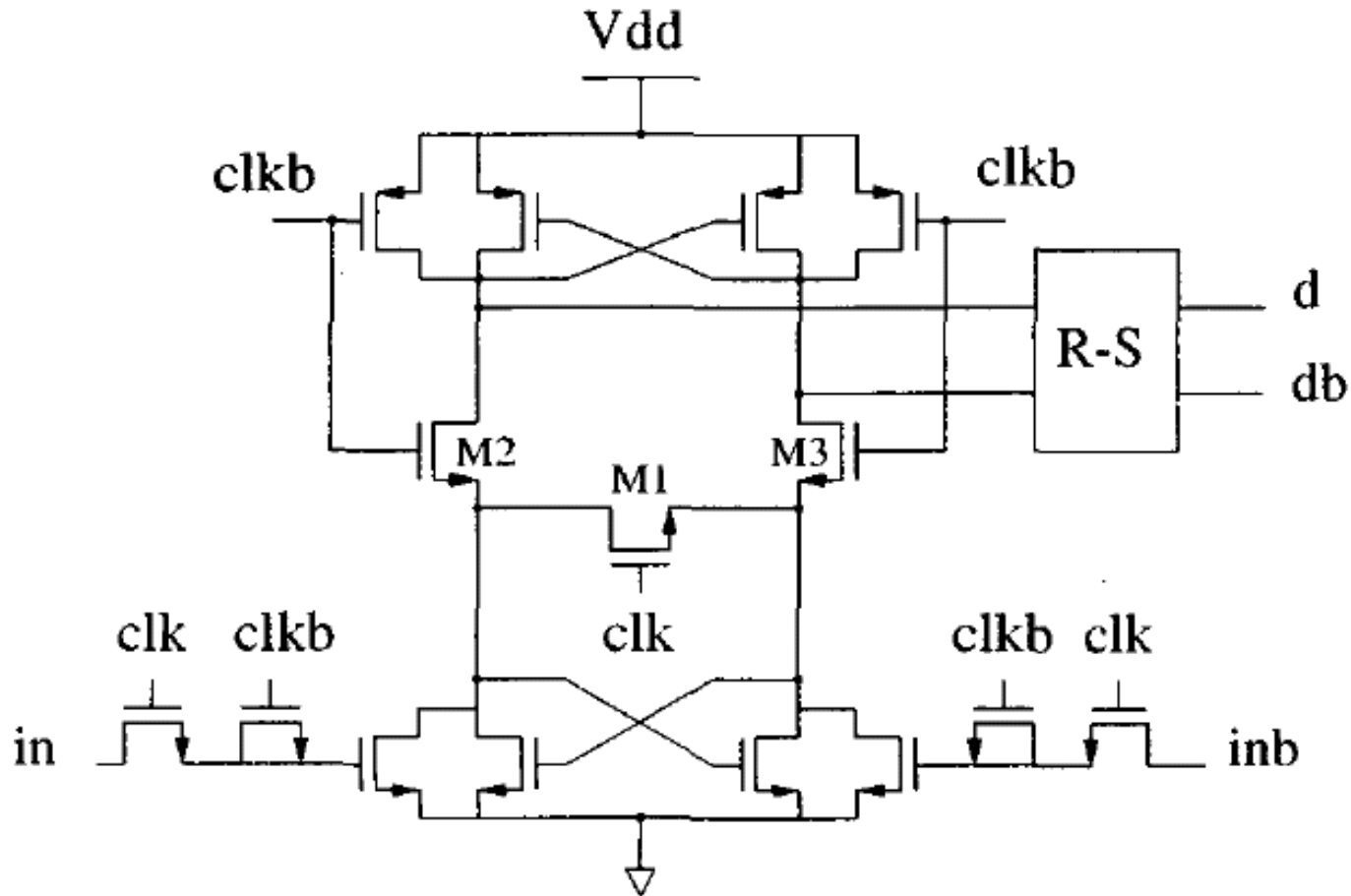
VLSI Rx Sampler Circuit Implementation



- NMOS for high side signal
- Diff pair for low offset
- Small signal operation
- High-speed operation
- Data keep in pre-charge



VLSI Rx Sampler Circuit Implementations



VLSI SC Rx Sampler/Eq Circuits

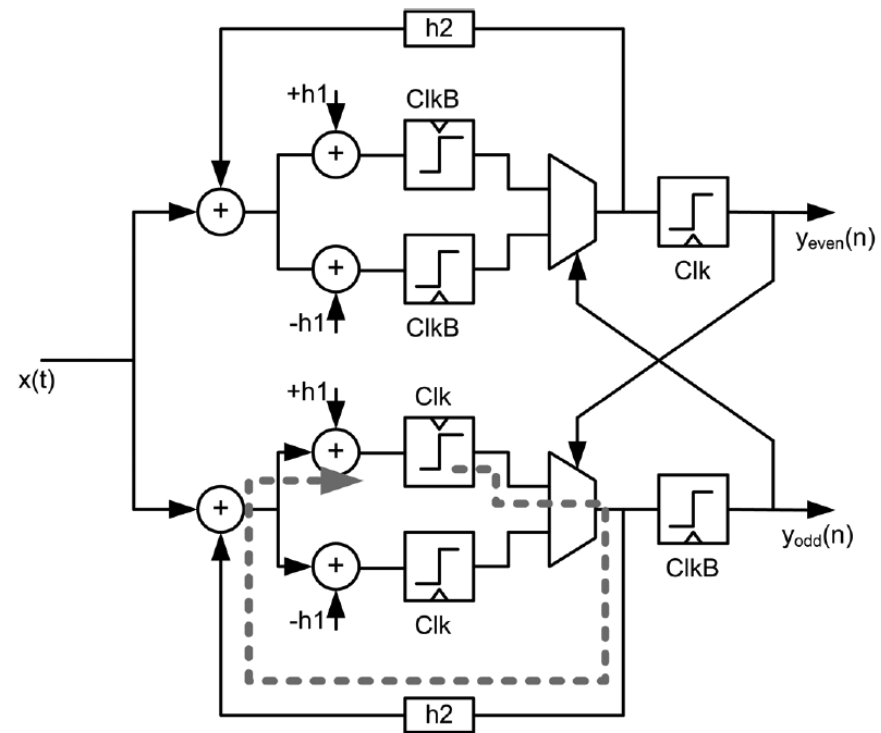
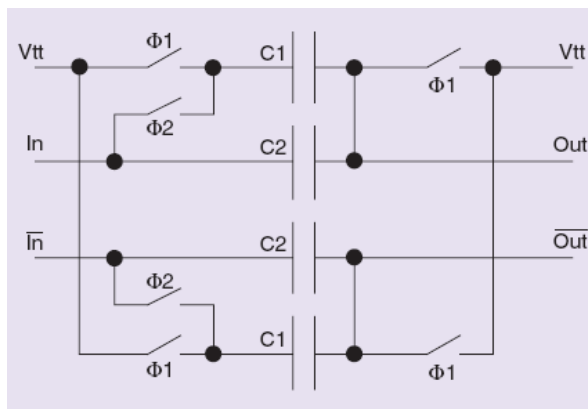
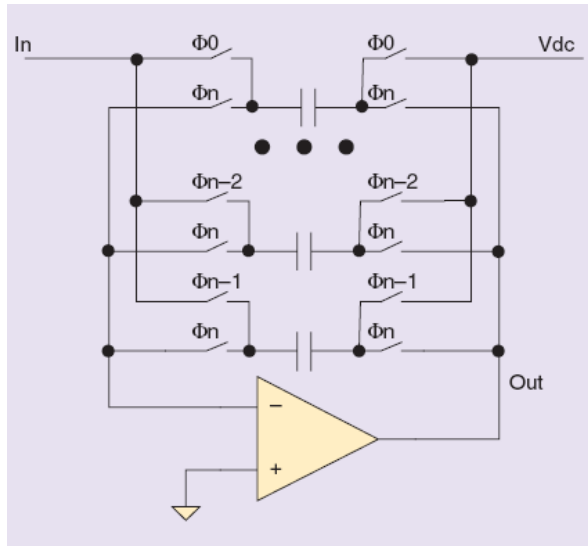
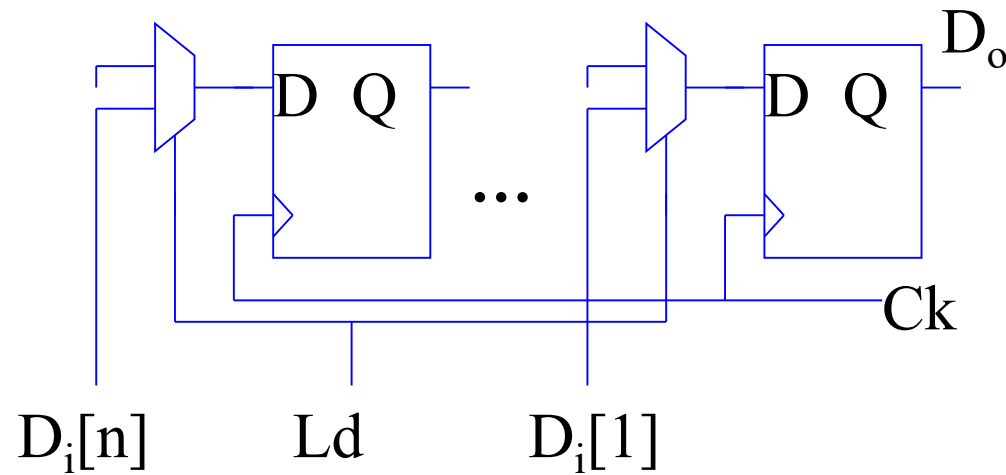
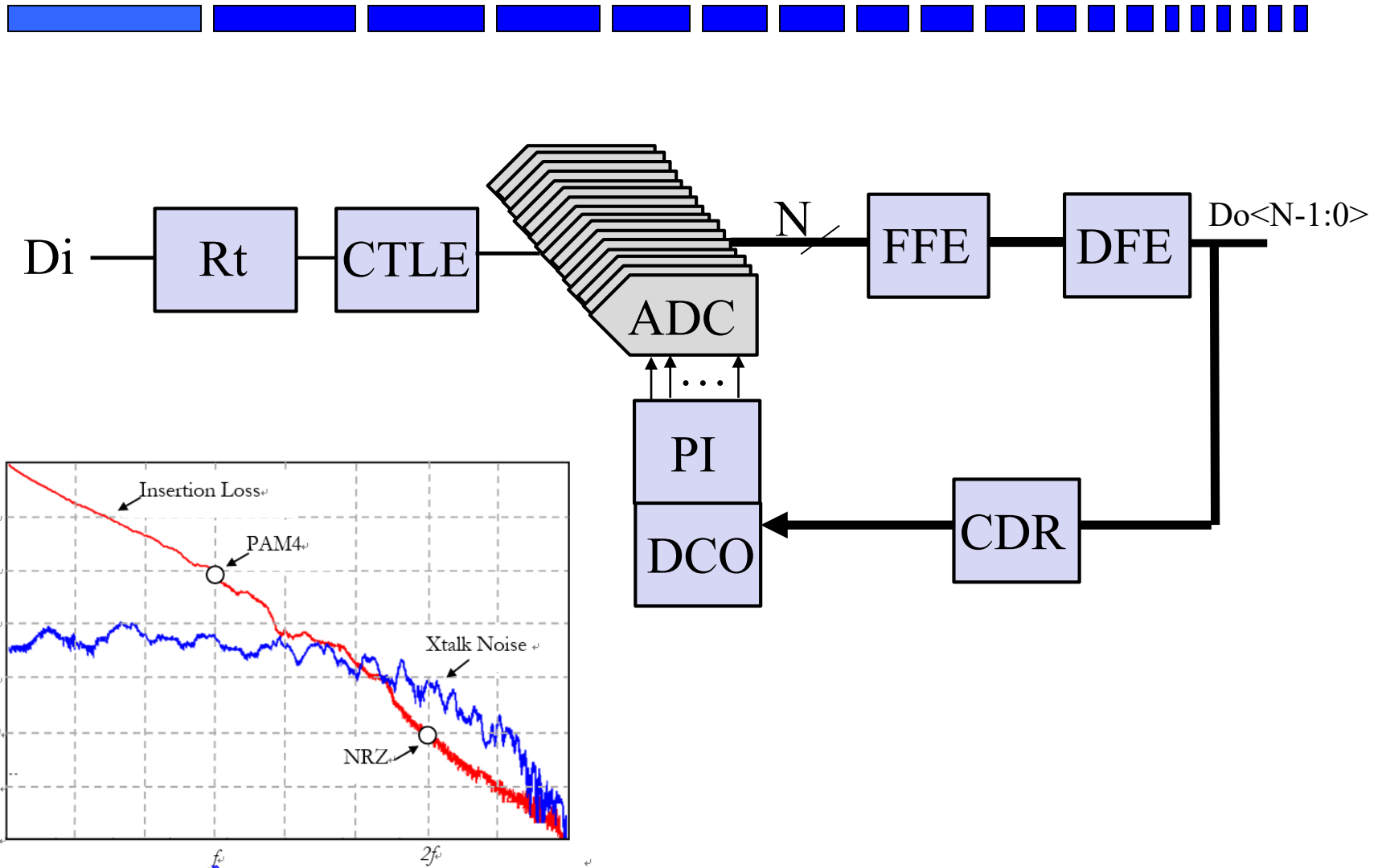


Fig. 2. Half-rate DFE receiver with speculation; new critical path is shown with the dashed line.

SIPO Circuit



PAM-4 Rx



Reading Assignment



- Please read Rx Chapter of the textbook